

Agenda

Dictionary and Sets

Tuples

Functions

List & Dict Comprehension

Lambda Functions

map, filter

modules & Packages

Builtin modules

File & Exception handling

3rd party modules

Dict	Tuple
{}	Similar to list
dict()	Immutable(cant be modified)
get()	index, count
clear()	indexing & slicing
copy()	
keys()	
values()	
items()	
pop()	

Which data structure will you use

1. 100 ip addresses to do a port scan
2. you have data about the host ip address & the open ports on it

I have a string, I want to count the number of characters in the string (freq counter)

vishwasksingh

v:1,i:2,s:3,h:2.....

Set

– Collection of unique items

Functions

```
demo_functions.py > ...
1  def fun_name():
2      pass
3
4  def display_greeting(nme):
5      print("Hello World", nme)
6
7  def add(a,b):
8      return a+b
9
10
11  display_greeting("Vishwas")
12  print(add(23,45))
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
C:\Users\VishwasKSingh\Workspace\ey-coh3-workspace>python demo_functions.py
Hello World Vishwas
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```

Structure of a script that we write (good Practice)

```
# Import Statements

# Constant Definitions

# Function Definitions

# Function Calling
```

In the global space you have to use the variables after you define it

In the local space when you just call the global variables, they are accessible

you can have a variable name(locally) same as variable name (globally), but priority is local variable value

If I want to modify a global variable inside a local function, I can define it as global

But not recommended for good practices

The file which is run directly is always assigned the `__name__` with a value `'__main__'`

In all other cases like imports (that file name is usually assigned)

`simple_interestv4.py`

```
def si(p1,p2,p3):  
    returns p1*p2*p3  
if __name__ == '__main__':  
    principal = input()  
    time_in = input()  
    rate = input()  
    print(si( ... ))
```

`test_si.py`

```
import simple_interestv4  
  
simple_interestv4.si(a,b,c)
```

Case 1:

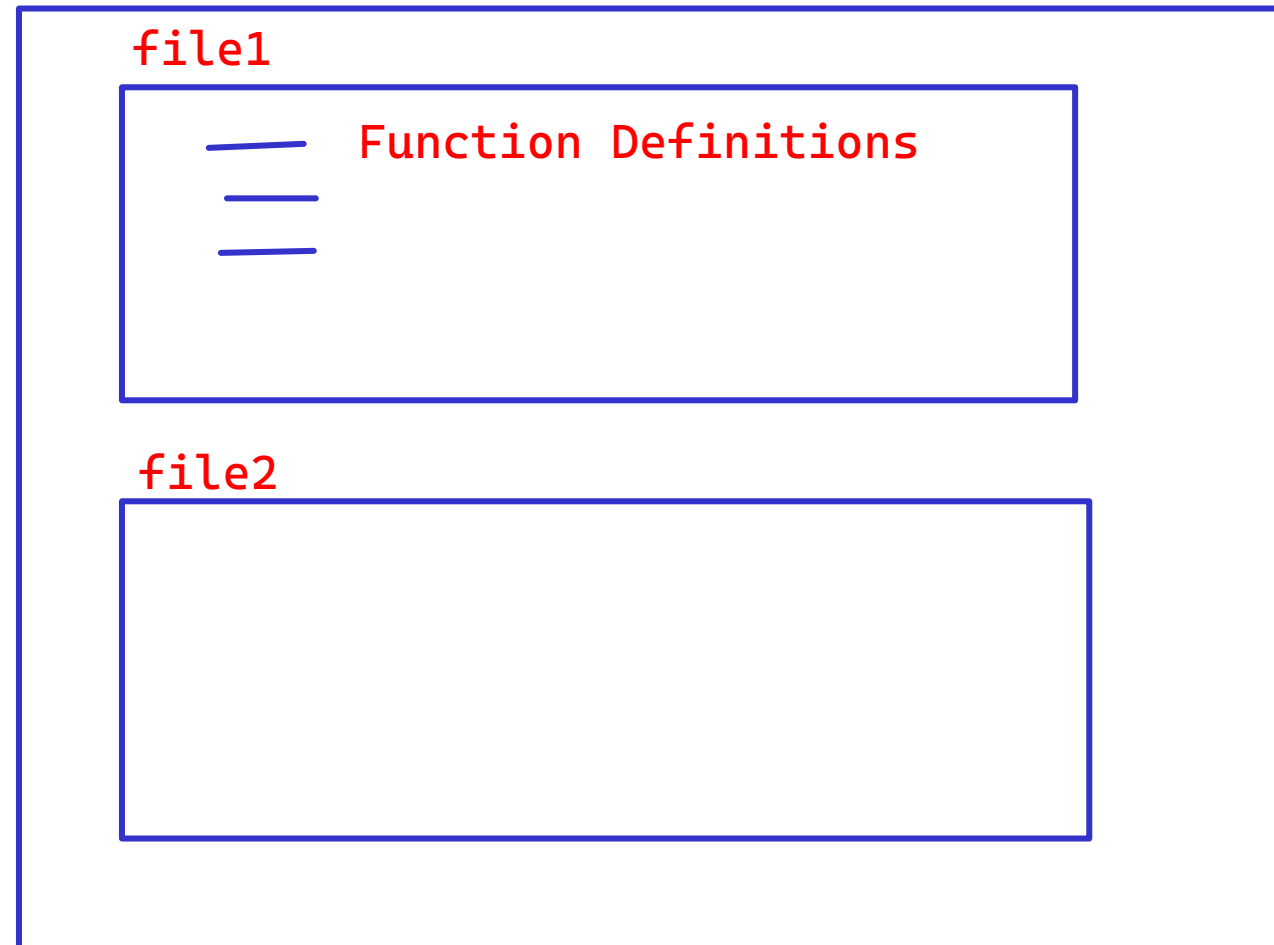
```
python simple_interestv4.py  
__name__ → __main__
```

case 2:

```
python test_si.py  
what will be value for test_si.py file  
__name__ → __main__
```

```
what will be value for simple_interestv4.py as it is imported  
__name__ → simple_interestv4
```


Folder



Efficient handling of lists and dictionaries

List Comprehension

I want a list of numbers from 1 to 1000

```
l1 = []  
for i in range(1,1001):  
    l1.append(i)
```

```
l1 = [i for i in range(1,10001)]
```

Assume you are a security analyst, You have IAM Dump with you. You are auditing privileges for users and their access level

```
user_roles = {  
    "vishwas.singh": "Admin",  
    "Arjun.reddy" : "User",  
    "Ganesh" : "User",  
    "John" : "Admin"  
}
```

Find the users with Admin access

lambda functions

short anonymous functions

```
lambda v: x**2
```

Ditionary Comprehension

```
d1 = {i:i**2 for i in range(1,11)}
```

You have a dictionary of users and their failed attempt count, find the users who have maximum failed attempts

```
user_attempts = {  
    "vishwas":2,  
    "shruti":5,  
    "John":9  
    "Riya":16  
    "Ganesh":1  
}
```

```
[("vishwas",45,92),("arjun",23,6),("John",1,2)]
```